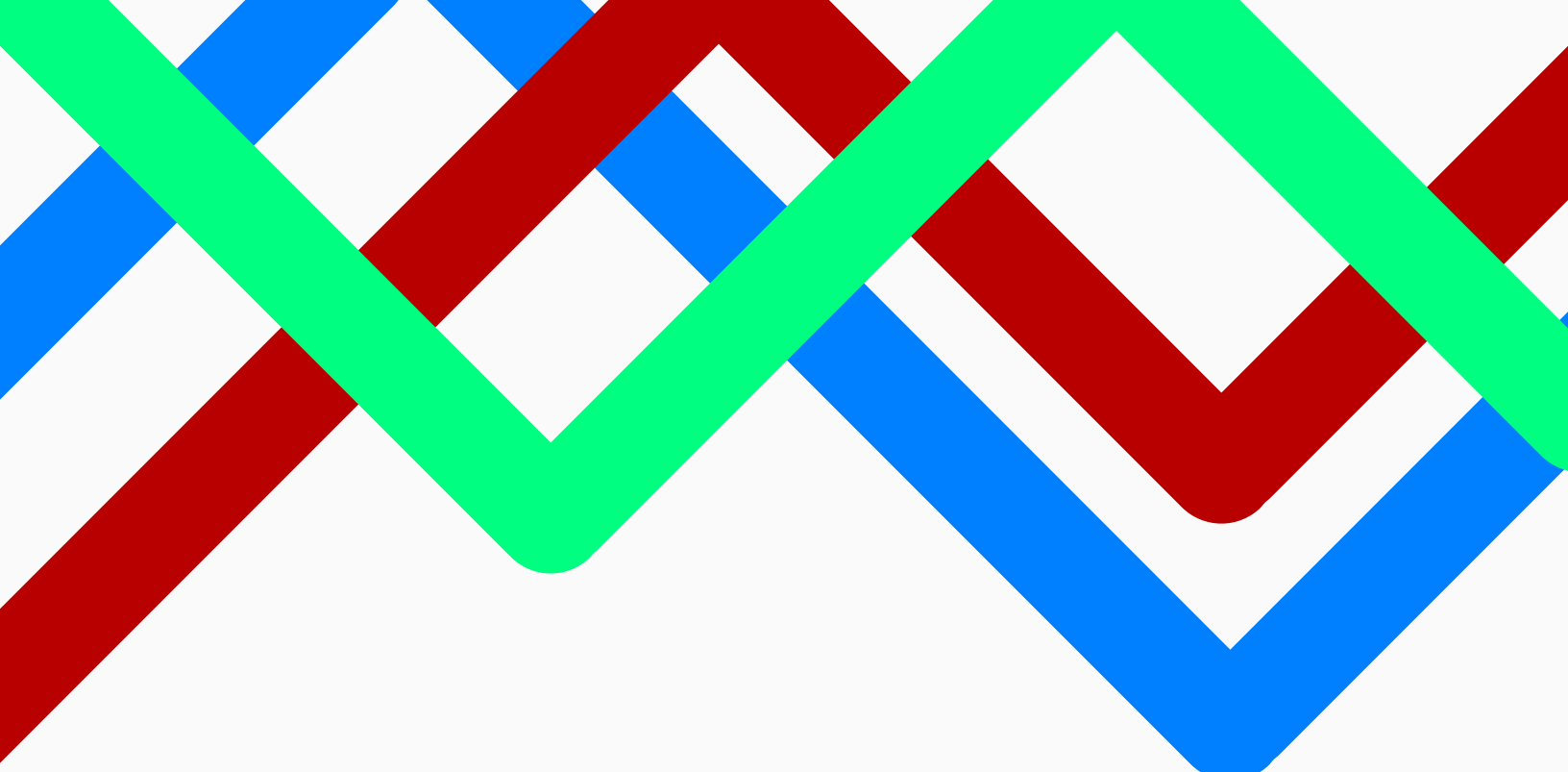




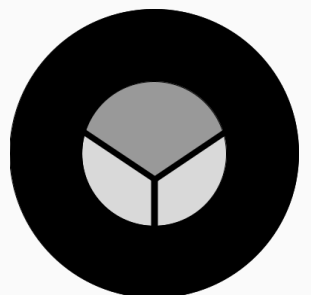
Cracker Protein Lab

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DE Intro to Biology I

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“Truth is ever to be found in the simplicity, and not in the multiplicity and confusion of things” - Issac Newton

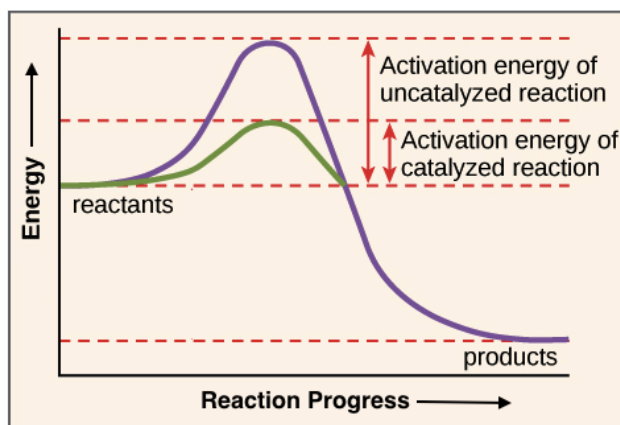


1 Abstract

To show the denaturing of amylase through the change in pH levels and how the efficacy and efficiency of enzyme performance are scale-based.

2 Background / Introduction

An enzyme is a catalyst that speeds up the rate of a reaction. An enzyme is not destroyed during this reaction and can be used multiple times. Enzymes work by lowering activation energy of a catalyst reaction.



Enzymes have four factors that affect rate of reaction. Each specific enzyme works best at specific temperatures and pH levels. Generally, rising temperature speeds up a reaction, but extreme heat may cause an enzyme to denature. Additionally, each enzyme has an ideal pH range. Extreme pH values may cause enzymes to denature. Denaturing occurs when the enzyme loses its shape due to these stressing factors. This lab explores denaturing of a specific enzyme, amylase, via acidity.

Amylase is an enzyme that helps with the digestion of carbohydrates. It is made in the pancreas and in saliva glands. Amylase is categorized into "3 main classes -- alpha-, beta-, and gamma amylases-- each targeting distinct segments of the carbohydrate molecule. Alpha amylase is present in humans, animals plants and microbes, whereas beta amylase is primarily found in microbes and plants. Gamma amylase, on the other hand, can be found in both animals and plants." **CITATION NEEDED** Amylase also stays stable for 4 days at room temperature, so there is no worry about spoiling of amylase.

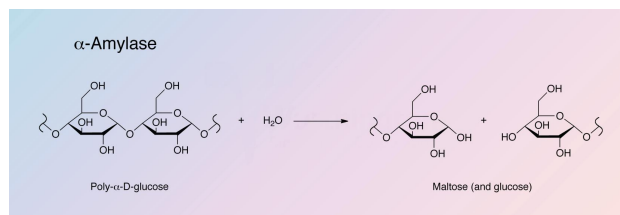
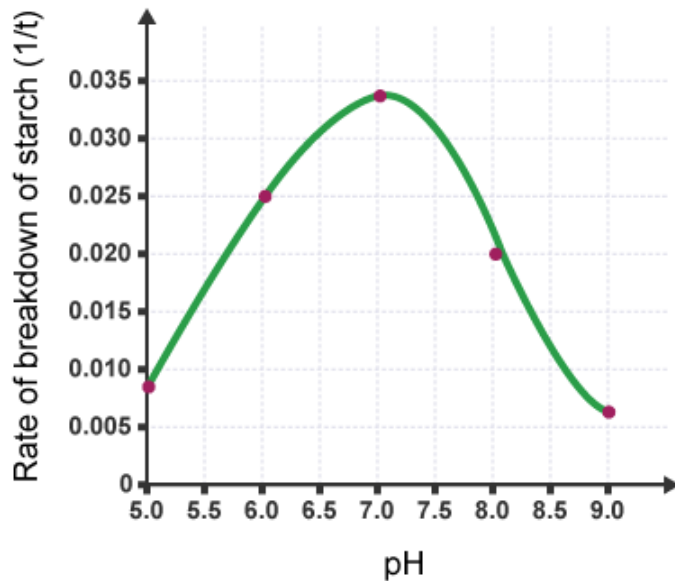
α amylase has an optimal pH of 7, following this curve.

α -amylase splits linear carbohydrate polymers at internal bonds. **CITATION NEEDED** It does so via the process of hydrolysis. Considered the reverse of the polymer building process dehydration synthesis, "(hydrolysis) is a chemical process in which a molecule is cleaved into two fragments by the addition of a molecule of water. One fragment of the parent molecule gains a hydrogen ion (H^+), while the other group collects the remaining hydroxyl group (OH^-)." **CITATION NEEDED** Amylase completes this by cleaving a starch (a polysaccharide) into "smaller molecules, ultimately yielding maltose, which in turn is cleaved into two glucose molecules." **CITATION NEEDED**

This correlation between the presence of amylase in saliva and

3 Question / Problem

How does *amylase*, an enzyme found in saliva, break polysaccharides into simpler carbohydrates in different pH levels?



4 Elements & Hypothesis

4.1 Hypothesis

If the pH levels of three solutions of pH {4, 5.5, 7}, are exposed to the enzyme *amylase* found in saliva and mixed with a polysaccharide containing cracker, then there should be a trend whereas the pH 7 solution will have the highest density of glucose, the pH 5.5 solution should have the medium density of glucose, and the pH 4 solution should have the lowest density of glucose.

4.2 Experimental Elements

- Independent Variable: PH Level of the solution
- Dependent Variable: Glucose concentration produced (in mmol/L)
- Experimental Groups:
 - Group 1: Saltine Cracker, Saliva, pH 4.0 solution
 - Group 2: Saltine Cracker, Saliva, pH 5.5 solution
- Positive Control Group:
 - Saltine Cracker, Saliva, optimal pH solution (pH 7.0)
- Negative Control Group:
 - Saltine Cracker, optimal pH solution (pH 7.0)

- Controls
 - Saliva to Solution ratio
 - Cracker to Solution ratio
 - Time allowed for enzymatic processes
 - Method of glucose measurement

5 Materials / Set up

5.1 Materials

The following materials were gathered:

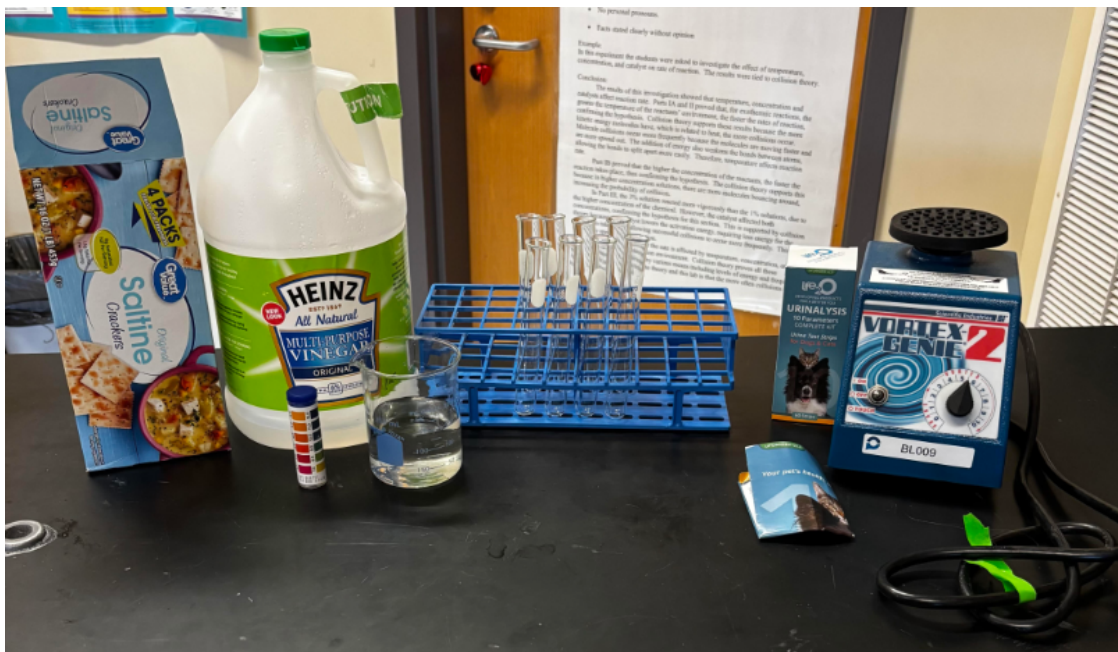
1. 4 saltine cracker pieces (approximately 1 gram each)
2. Roughly 100 mL of Heinz all-natural vinegar
3. pH test strips
4. Roughly 250 mL of distilled water
5. 8 15-25mL test tubes
6. Test tube rack
7. Life2O Pet Urinalysis Test Strips: Dog and Cat Urine Test Strips
8. Genie 2 Vortex Mixer SI-0236
9. Fresh Human Saliva (Roughly 20 mL total)
10. Stopwatch
11. Mortar & Pestle
12. 4 stirring rods
13. Plastic gloves
14. Tape
15. Black Marker
16. 2x 250 mL beaker
17. Plastic test tubes

5.2 Set Up Procedure

1. All lab materials were collected and thoroughly inspected to ensure cleanliness.
2. Two sets of four test tubes were labeled with {"C","pH=7","pH=5.5","pH=4"} using tape and a marker.
3. The first set of four test tubes were filled with 10 mL of distilled water.
 - The water was filled up in the first beaker with about 100 mL of distilled water.
 - The vinegar was filled up to about the 50 mL mark of the second beaker.

- Then, according to the test tube labels, the first set of four had 2 mL of vinegar added incrementally (until the pH was correct) and checked via a pH strip.
- When all four test tubes of the first set were at the pH level indicated by each respective label, the tubes were put away, in a row, back onto the test tube rack
- All lab members put on their gloves.

An exemplar image of an example setup without pH solutions can be found below:



6 Methods / Procedure

- Saliva was collected by each individual labmate into their respective plastic test tube. This was to minimize the safety impact of having different contaminants on each test tube.
- Four crackers (approximately 1 gram each) were crushed by a mortar and pestle into a fine powder.
- The four crackers were distributed evenly across the **second** set of test tubes. (So now, the second set only has cracker powder in it.)
- The saliva was mixed into a fourth plastic test tube and vortexed via the Genie 2 Vortex Mixer
- The labmates were careful here; the saliva was then distributed evenly into the **first** set of test tubes. (So now, the first set only has solution + saliva. Only $\frac{3}{4}$ contained saliva because of the negative control.)
- Carefully, the first set of test tubes was added onto each respective test tube of the second set and began a stopwatch.
- This is where the rush began. During the 30 seconds of curing (determined via the stopwatch), each test tube was vortexed to ensure that each solution was at least somewhat homogenous. This was a step where caution was taken to ensure that mixture escaped the test tube.
- After the 30 seconds of curing (determined via the stopwatch) was over, one urinalysis strip was placed into each test tube to allow for glucose measurement.

9. The stopwatch was reset and began counting to 30 seconds again. After the 30 seconds were over, each urinalysis strip was carefully removed from each test tube. Then, the urinalysis strip was compared to the package, and the glucose levels were measured and recorded.

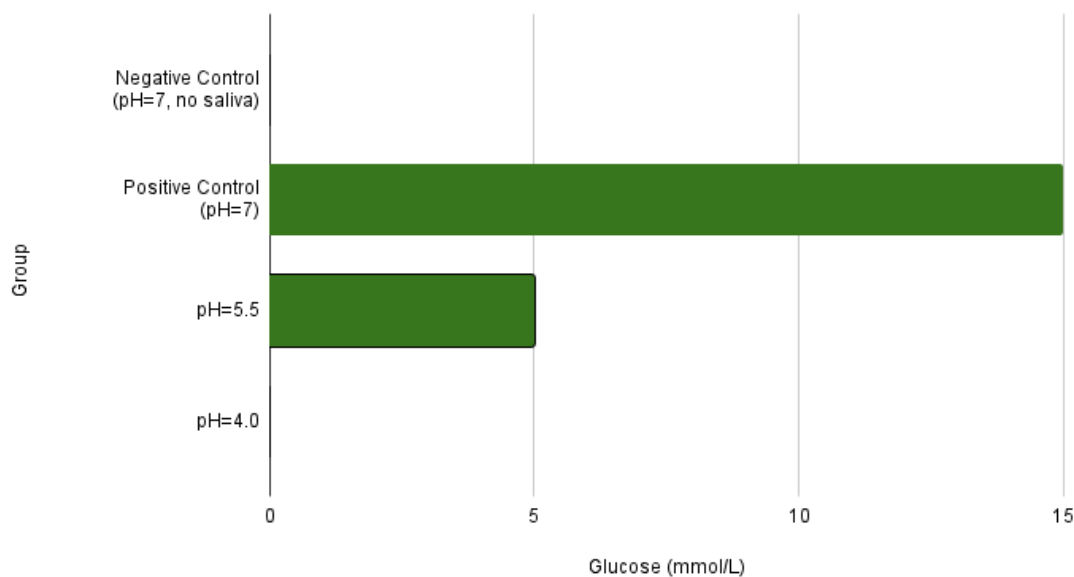
7 Data / Results

7.1 Table

Group	Glucose Measured (mmol/L)
Negative Control (pH=7.0, NO saliva)	0
Positive Control (pH=7.0)	15
pH=5.5	5
pH=4.0	0

7.2 Graph

Glucose (mmol/L) versus Group



8 Conclusion / Discussion

The lab results support the hypothesis. Both control groups (Positive, Negative) behaved as expected. Since the negative control does not contain saliva, it shouldn't have amylase. This removed sources of possible error from faulty urinalysis strips. Additionally, the positive control group behaved as expected, resulting in the highest concentration of glucose. Since the optimal pH level for amylase is 7.0 **INSERT CITATION HERE**, it is within reason to correlate the two. The two experimental groups behaved as expected. The acidic pH 4.0 solution had similar results to the negative control because at a pH of 4.0, amylase denatures **INSERT CITATION HERE**. The less acidic pH 5.5 solution also behaved as expected, yielding in a result greater than 0 but less than the positive control group. This, hence, means that the hypothesis can be accepted.